

PPT "RLGC Design"

1. Cylindrical TSV why?

2. Types of Networks

3. Types of Models.

4. Defect free GS TSV

$\left. \begin{matrix} C \\ L \\ R \\ G \end{matrix} \right\}$ They considered
 π model.

5. Defected GS TSV (pin hole)

$\left. \begin{matrix} C \\ L \\ R \\ G \end{matrix} \right\}$ proposed
 model.

Resistance defect free:

→ Refer to 30/05/26 [Day 04]
 or working for resistance
 concept.

Conductance defect free:

→ Refer to 30/05/26 [Day 04]
 or working for Conductance
 Concept.

01 [Working on]: "Defect free & Defected
 TSV modelling electric-
 al"
 [High scalable paper]

01 Defect free

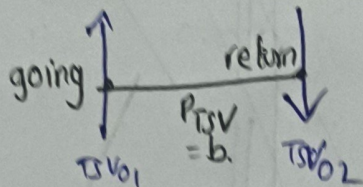
RLGC model represents the
 parasitic parameters of via.

(1/m units)

1.1 A defect free TSV assumes perfect
 geometry & dielectric material

$$R = \frac{\rho l}{A} \quad \begin{matrix} \rho = \text{resistivity of con.} \\ l = \text{TSV height} \\ r = \text{TSV radius.} \end{matrix}$$

$$L = \frac{\mu h}{2\pi} \ln\left(\frac{b}{a}\right) \quad \begin{matrix} h = \text{TSV height} \\ a = \text{TSV radius} \\ b = \text{reference} \\ \text{dis. for return} \\ \text{path radius. to} \\ \text{signal path} \end{matrix}$$



$$C = \frac{2\pi \epsilon h}{\ln\left(\frac{b}{a}\right)}$$

$$G = \frac{2\pi \sigma_d h}{\ln\left(\frac{b}{a}\right)} \quad \sigma_d = \text{dielectric} \\ \text{conductivity} \\ \text{of TSV.}$$

02 Defected TSV RLGS Equations:

1.1 Resistance of cu almost
 unchanged. $R_0 \approx R_{\text{pinhole}} = \frac{\rho l}{A}$

1.2 Inductance varies a little

$$L_{\text{pinhole}} \approx L_0 = \frac{\mu h}{2\pi} \ln\left(\frac{b}{a}\right)$$

$$1.3 \quad C_{\text{pinhole}} \approx \frac{2\pi \epsilon h}{\ln\left(\frac{b}{a}\right)} \approx C_0$$

1.4 Conductance.

$$G_0 = \frac{2\pi \sigma_d h}{\ln\left(\frac{b}{a}\right)}$$

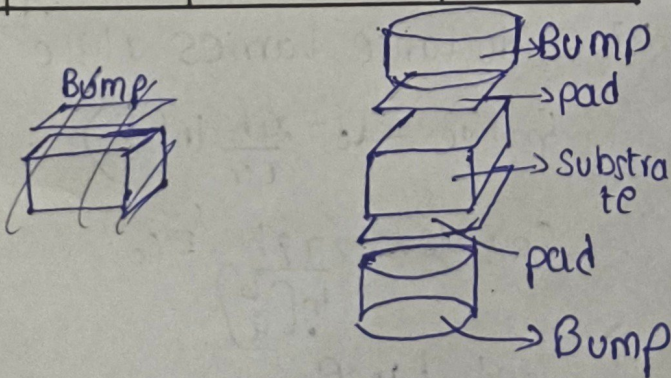
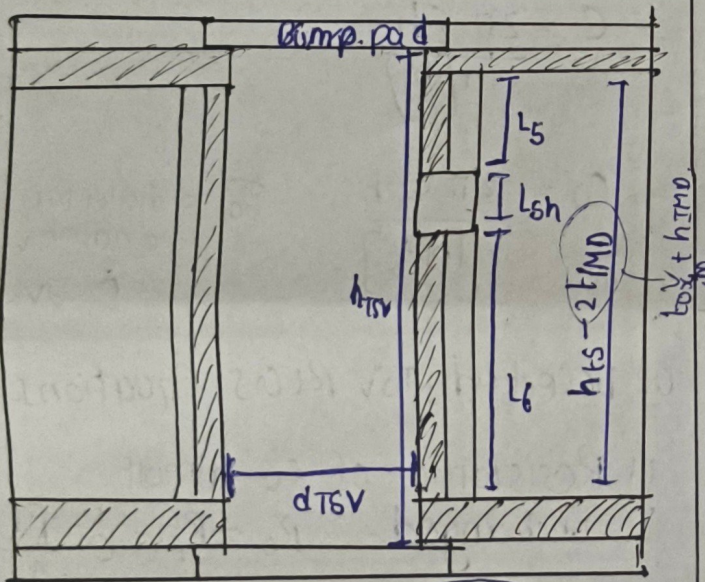
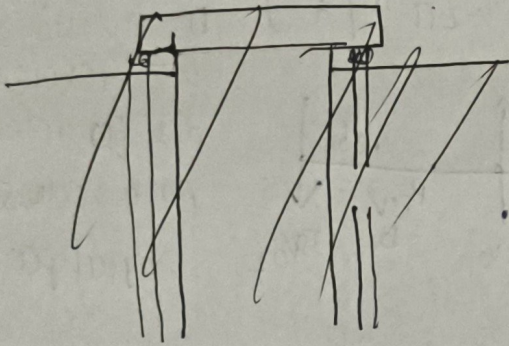
$$G_{\text{pinhole}} = G_0 + G_{\text{leak.}}$$

{ OR }

- next page

Defected TSV modelling
 → RL pin hole design.

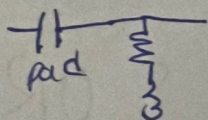
Crosstalk in Coupled transmission lines
 [Modelling & simulation in HFSS]



Bump X

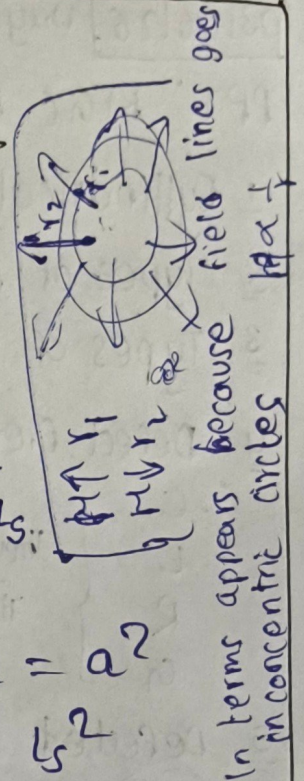
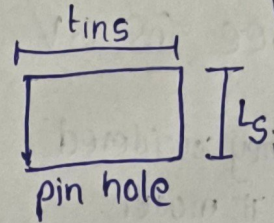
pad

Si Sub. ins pad



$$\frac{1}{\text{shunt resistance}} = \frac{\int_T t_{ins}}{L_s^2}$$

$$R = \frac{\rho l}{A}$$



l_s square so area = a^2
 L_s^2

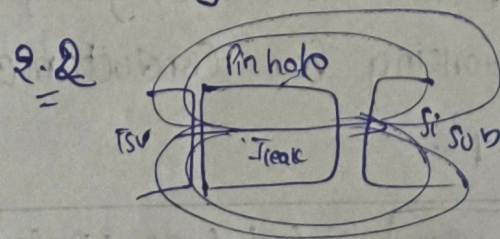
$$L_{sht} = \frac{\mu_0 \mu_{tsv}}{2\pi} t_{ins} \left[\cosh^{-1} \left(\frac{t_{ins}}{\sqrt{L_s^2 + t_{ins}^2}} \right) \right]$$

general. complex (in term)

$$- \sqrt{2 \cdot t_{ins}^2 + L_s^2} + \frac{t_{ins}}{a} + \sqrt{L_s^2 + t_{ins}^2}$$

correction factor.

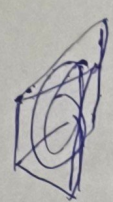
2.1 inductance associated with leakage current path through the pin hole.



$$L = \frac{\mu l}{2\pi} \left[\ln \left(\frac{l}{r} \right) - 1 \right]$$

$$3 \quad G_{leak} = \frac{2\pi\epsilon_0\epsilon_{ox} [l_{TSV} - jr_{TSV}]}{\ln\left(\frac{r_{TSV}+t_{ox}}{r_{TSV}}\right)}$$


3.1 general cylindrical capacitor.

$$C = \frac{2\pi\epsilon l}{\ln\left(\frac{b}{a}\right)}$$


3.2 l becomes $l_{effective}$.

$$l_{TSV} - jr_{TSV} = \rho$$

angular extent of pinhole.

surface area

where

$$j = \frac{\omega h_{pin}}{2\pi r_{TSV}} \frac{c\omega l}{A}$$

$$j = \frac{\text{damaged area}}{\text{sidewall area}}$$

TSV sidewall affected by pinhole.

$$4 \quad G_{leak} =$$

$$B.1 \quad G = \frac{2\pi\sigma h}{\ln\left(\frac{b}{a}\right)}$$

cylinder,

circumference.

$$4.2 = \frac{2\pi\epsilon_{SiO_2} l_{TSV} - 4\pi^2 j f \epsilon_0 \epsilon_{air} r_{TSV}}{\ln\left(\frac{r_{TSV}+t_{ox}}{r_{TSV}}\right)}$$

oxide leakage contribution

— Pinhole correction

Radical field spreading.

01 Pinhole shape in RL model

02 materials they used for pinhole.

we are keeping.

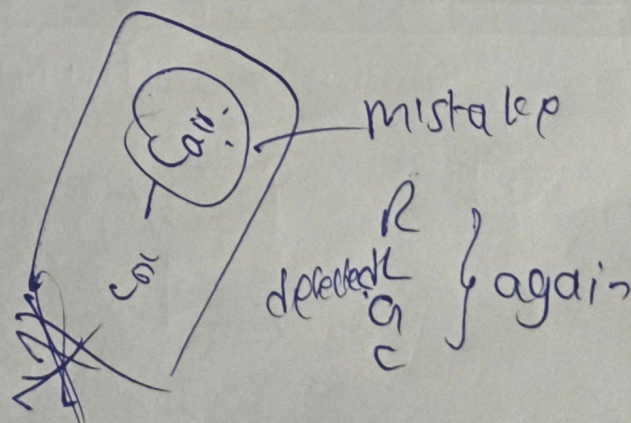
03 f_{air} search.

$$G_{leak} = \frac{2\pi\epsilon_{rox}\epsilon_0 [l_{TSV} - jr_{TSV}]}{\ln\left(\frac{r_{TSV}+t_{ox}}{r_{TSV}}\right)}$$

$$= \frac{2\pi\epsilon_0\epsilon_{rox}}{\ln(x)} - \frac{2\pi\epsilon_0\epsilon_{rox} [jr_{TSV}]}{\ln(x)}$$

$$= \dots = 2\pi\epsilon_0\epsilon_{rox} \left(\frac{\omega h_{pin} r_{TSV}}{2\pi r_{TSV}} \right)$$

$$= \omega h_{pin} \epsilon_0 \epsilon_{rox}$$



04 video: Vijaya Rao kumbhare interconnects

ALU digital academy
lecture: 5. (interconnects)